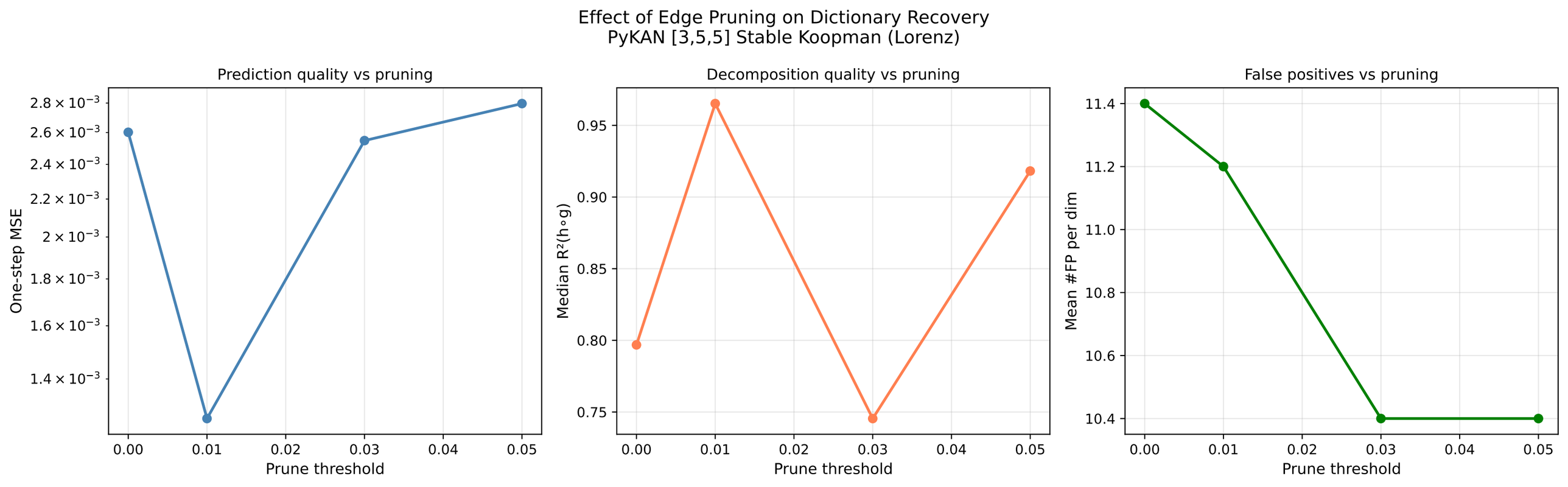
Effect of Edge Pruning on Dictionary Recovery
PyKAN [3,5,5] Stable Koopman (Lorenz)
Prediction quality vs pruning
One-step MSE
2.8×10^{-3}
2.6×10^{-3}
2.4×10^{-3}
2.2×10^{-3}
2×10^{-3}
1.8×10^{-3}
1.6×10^{-3}
1.4×10^{-3}
0.00
0.01
0.02
0.03
0.04
0.05
Prune threshold
Decomposition quality vs pruning
Median R²(h∘g)
0.95
0.90
0.85
0.80
0.75
0.00
0.01
0.02
0.03
0.04
0.05
Prune threshold
False positives vs pruning
Mean #FP per dim
11.4
11.2
11.0
10.8
10.6
10.4
0.00
0.01
0.02
0.03
0.04
0.05
Prune threshold